

Water

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 94-264

WASTE DISCHARGE REQUIREMENTS

FOR
COUNTY OF SAN JOAQUIN
COUNTY SERVICE AREA NO. 15 WWTP
SAN JOAQUIN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The County of San Joaquin (hereafter Discharger) submitted a Report of Waste Discharge, dated 23 February 1994, for their wastewater treatment and disposal facility. The property (Assessor's Parcel No. 087-100-58) is owned by the County of San Joaquin.
2. Waste Discharge Requirements Order No. 79-15, adopted by the Board on 26 January 1979, prescribes requirements for a discharge from the facility to land.
3. Order No. 79-15 is neither adequate nor consistent with current plans and policies of the Board.
4. The County Service Area No. 15 Wastewater Treatment Plant was originally constructed in 1966 and was relocated to the present site in 1976 to serve highway services and limited industrial area east of the State Highway 99 along Waterloo Road (State Highway 88), east of the City of Stockton. The facility has not reached its design capacity.
5. The Report of Waste Discharge describes the proposed discharge as follows:

Monthly Average Flow: 0.08 mgd
Daily Peak Flow: 0.40 mgd
Design Flow: 0.20 mgd

<u>Constituent</u>	<u>Influent</u>	<u>Effluent</u>
BOD ₅	298 mg/l	20 mg/l
TSS	180 mg/l	34 mg/l

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6. The County Service Area No. 15 Wastewater Treatment Plant proposes to discharge 0.08 mgd of monthly average flow from commercial and light industry wastewater to land. The monthly average flow of 0.08 mgd includes existing flow plus a 33% flow increase through annexation. The facility utilizes a package treatment plant which consists of an activated sludge process with an aerated contact tank followed by a concentric ring clarifier. Chlorination is employed prior to discharging into the percolation/evaporation ponds. A fourth pond is used as a sludge drying bed. The flow is rotated among the ponds to allow for mosquito control and discing of the ponds to control the weed growth. Ground water in this area occurs at 90 feet below the ground surface with a south-westerly flow.
7. On 30 October 1991, West Yost and Associates, an engineering consultant retained by the County, submitted an evaluation and expansion study on the County Service Area No. 15 WWTP. The study notes that with a peaking factor of 3.25, the maximum short duration that should be allowed based on clarifier capacity was estimated to be at 0.405 mgd and average rated capacity of 0.125 mgd on the existing wastewater treatment facility. The original rated capacity design of 0.20 mgd was assumed to be calculated on a minimal peaking factor by the manufacturer, in 1966. Additionally, the Discharger lacks the ability to shut the facility for routine maintenance. The County's long-term intent for this facility is to annex the outlet to the City of Stockton's Regional Wastewater Treatment Facility.
8. The County Service Area No. 15 generates approximately 600 ft³ of dry sludge annually. The sludge in the designated drying pond is incorporated into the soil by discing annually.
9. The facility is in a flood zone according to the Federal Emergency Management Agency's Flood Insurance Rate Map. This zone can experience a 100 year flood to a depth of less than one foot. The berm surrounding the ponds is elevated four feet above the adjacent grade and will protect the ponds from a one foot flood depth.
10. San Joaquin County Service Area No. 15 is in Section 30 , T2N , R6E, MDB&M, with surface water drainage to Calaveras River, as shown in Attachment A, which is attached hereto and part of the Order by reference.
11. The Board adopted a Water Quality Control Plan, Second Edition, for the Sacramento - San Joaquin Basin (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

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12. The beneficial uses of Calaveras River are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
13. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
14. San Joaquin County has adopted a Negative Declaration in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.), and the State Guidelines. This project, as approved, will not have a significant effect on water quality.
15. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
16. This discharge is exempt from the requirements of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15). The exemption, pursuant to Section 2511(b), is based on the following:
 - a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to 22 CCR, Division 4, Chapter 30, as a hazardous waste.
17. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
18. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 79-15 is rescinded and the County of San Joaquin, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

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A. *Discharge Prohibitions:*

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

B. *Discharge Specifications:*

1. The monthly average dry weather discharge flow shall not exceed 0.125 million gallons/day.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.
3. As a means of discerning compliance with Discharge Specification No. 2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
5. Ponds shall not have a pH less than 6.5 or greater than 8.5.
6. Ponds shall be managed to prevent breeding of mosquitos. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

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7. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
8. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Freeboard shall never be less than two feet measured vertically to the lowest point of overflow.
9. On or about **1 October** of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification 8.

C. *Sludge Disposal:*

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15, Division 3, Title 23, of the California Code of Regulations and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
3. Use and Disposal of sewage sludge shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*.

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D. Ground Water Limitations:

The discharge shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified.
2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
4. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 94-264, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

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4. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
6. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
7. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 September 1994.



WILLIAM H. CROOKS, Executive Officer

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Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 94-264
FOR

COUNTY OF SAN JOAQUIN
COUNTY SERVICE AREA NO. 15 WWTP
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INFLUENT MONITORING

The following shall constitute the influent monitoring program:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD ₅	mg/l	Grab	Monthly
Suspended Solids	mg/l	Grab	Monthly
Settleable Solids	mg/l	Grab	Monthly
Flow Rate	mgd	Cumulative	Daily

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the disposal field. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of ponds will be considered adequately composited. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD ₅	mg/l	Grab	Monthly
Suspended Solids	mg/l	Grab	Monthly
Settleable Solids	mg/l	Grab	Monthly
TDS	mg/l	Grab	Quarterly
Nitrate as N	mg/l	Grab	Quarterly

POND MONITORING

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Free Board	Feet	-----	Daily

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SLUDGE MONITORING

A one time sampling of the sludge prior to wasting shall be performed for the following constituents in dry weight (mg/kg): Arsenic, Cadmium, Chromium, Copper, Lead, Mercury, Molybdenum, Nickel, Selenium, Silver, Zinc, Total Kjeldahl-N, Organic-N, Ammonia-N, Nitrate-N, Phosphorus, Potassium, pH, Total Solids (%). The results shall be submitted to the Regional Board 60 days after the adoption of this Order.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with the waste discharge requirements.

Monthly monitoring reports shall be submitted to the Regional Board by the **15th day** of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

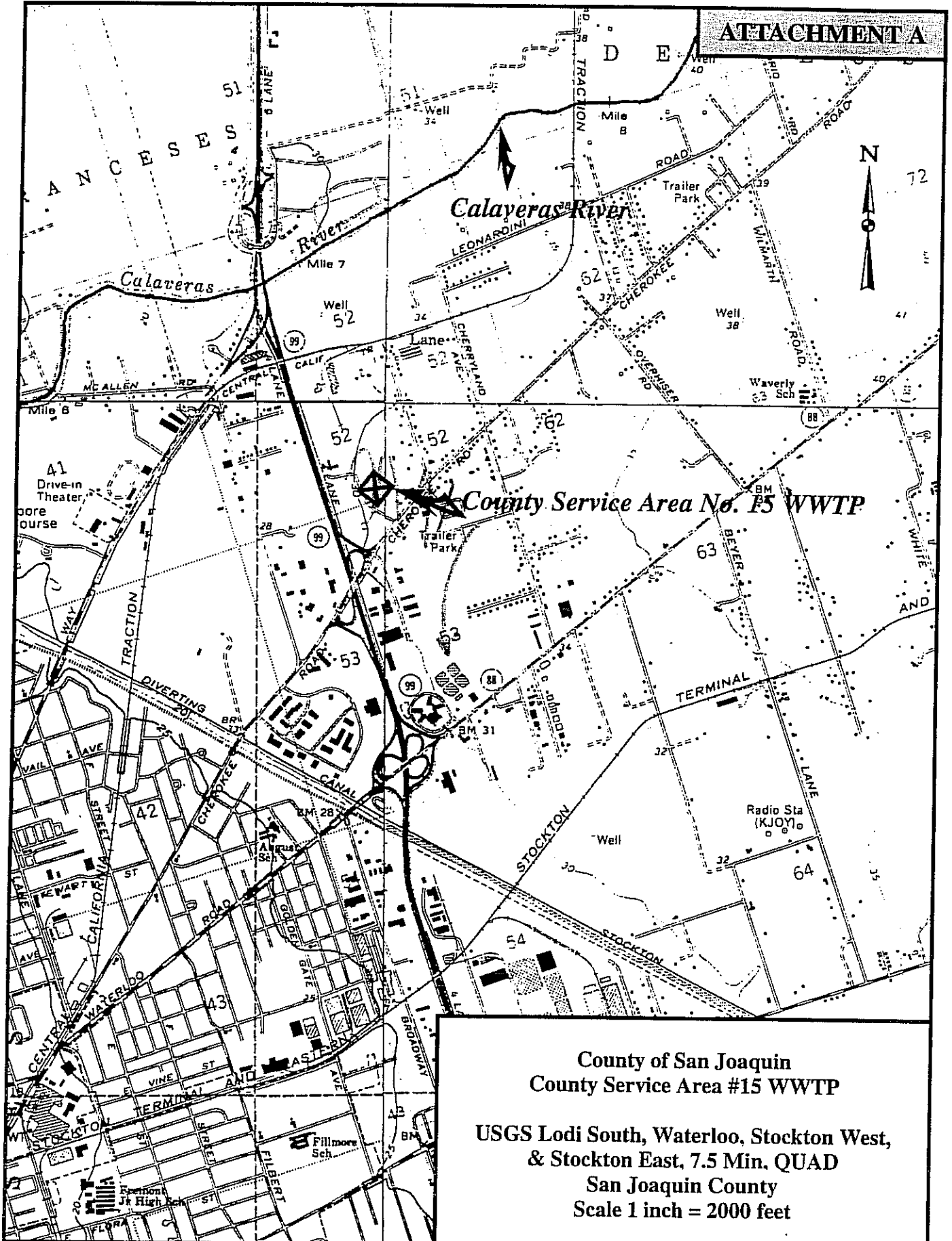
Upon written request of the Board, the Discharger shall submit a report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the Discharger into full compliance with the waste discharge requirements.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: William H. Crooks
WILLIAM H. CROOKS, Executive Officer

16 September 1994
(Date)

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INFORMATION SHEET

COUNTY OF SAN JOAQUIN
COUNTY SERVICE AREA NO.15 WWTP
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The County of San Joaquin Service Area No. 15 Wastewater Treatment Plant was relocated to the present site in 1978 to serve highway services and limited industrial (fast-food restaurants, hotels, and warehouses) area east of the State Highway 99 along Waterloo Road, east of the City of Stockton. The County Service Area No. 15 WWTP proposes to discharge an average monthly flow of 0.08 mgd. The monthly average flow of 0.08 mgd includes existing flow plus a 33% flow increase through annexation. The discharge consists of commercial and light industry to land. The wastewater treatment facility was rated at 0.125 mgd design capacity by a consulting firm (West Yost & Associates) retained by the County.

The facility utilizes a package treatment plant consists of an activated sludge process with an aerated contact tank followed by a concentric ring clarifier. Chlorination is employed prior to discharging into the percolation/evaporation ponds. A fourth pond is used as a sludge drying bed. The flow is rotated among the ponds to allow for mosquito control and discing of the ponds to control the weed growth. Ground water in this area occurs at 90 feet below the ground surface with a south-westerly flow.

The County Service Area No. 15 generates approximately 600 ft³ of dry sludge annually. The sludge in the designated pond is air dried prior to incorporating into the soil through discing.

The facility is in a flood zone according to the Federal Emergency Management Agency, Flood Insurance Rate Map. This zone can experience a 100 year flood to a depth of less than one foot. The berm surrounding the ponds is elevated four feet above the adjacent grade and will protect the ponds from a one foot flood depth.